Data-Quality Improvements and Applications of Long-Term Monitoring of Ionospheric Anomalies for GBAS

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ABSTRACT

An automated Long-Term Ionospheric Anomaly Monitoring (LTIAM) software package has been developed to support continuous ionospheric monitoring for the U.S. Local Area Augmentation System (LAAS) developed by the U.S. Federal Aviation Administration (FAA) in the Conterminous U.S. (CONUS). Continuous monitoring is needed to confirm the long-term validity of the ionospheric anomaly threat model [1,3] and update it if necessary. This is of particular importance over the next few years, as the intensity of solar storms is expected to peak in 2013-15. Prior work [2,4] has demonstrated that the LTIAM not only identifies extremely large ionospheric gradients but also supplies reliable ionospheric gradient statistics under all conditions. This tool can also be utilized to build ionosphere threat models for all regions where Ground-Based Augmentation Systems (GBAS) will be fielded in the future.

The LTIAM software enables the post-processing of data continuously collected by GPS reference station networks. Ionospheric gradients over short-baseline distances of 5 – 40 km can be observed using data collected from the Continuously Operating Reference Stations (CORS) network, which has over 1800 stations as of 2011 compared to about 400 stations prior to 2004. However, as the total number of stations increases, the number of stations with poor GPS data quality also increases. This is caused by the differing types of CORS receivers and antennas fielded by multiple organizations in various environments; some good, some not-so-good. The use of poor-quality data degrades the accuracy of ionospheric delay estimates and produces too many faulty anomaly candidates. Thus, station selection criteria need to be defined to reduce processing time in both automated procedures and manual analysis and validation while maximizing the benefit of this tool.

1.0. INTRODUCTION

TBC

2.0. LTIAM OVERVIEW

TBC

3.0. POOR CORS DATA QUALITY

This section investigates the effect of GPS data collected from stations with poor GPS data quality on the results of the LTIAM tool. Figure 1 shows the results from LTIAM with a threshold of 200 mm/km on a nominal day, 26 May 2012, during which geomagnetic storm condition was very quiet. On this particular day, it is expected that any severely large ionospheric gradients are not observed. However, the LTIAM tool returned many ionospheric anomalies because of the bad GPS data used. All 92 faulty threat candidates, marked with blue diamonds in ionospheric anomaly threat space, should be manually validated to confirm that these points are not real anomalies. If stations with poor GPS data quality are effectively removed by a methodology presented in this paper, the most of faulty candidates can be removed as shown in Figure 2. Therefore, we can save time and effort spending to manually validate the faulty candidates.

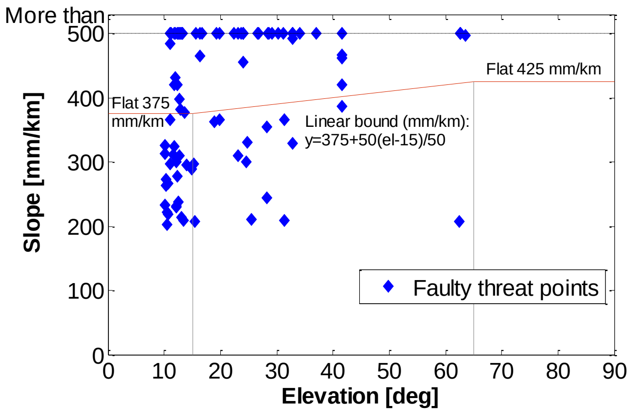

Figure 1. Faulty Candidates (92 points) Populated on Ionospheric Anomaly Threat Space Before Removing Stations with Poor GPS Data Quality on 26 May 2012.

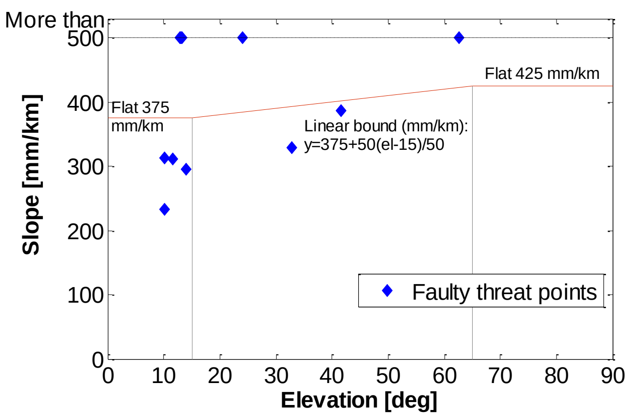

Figure 2. Faulty Candidates (11 points) Populated on Ionospheric Anomaly Threat Space After Removing Stations with Poor GPS Data Quality on 26 May 2012.

Figure 3 shows the slant ionospheric delays observed from two nearby stations OKEE and AVCA while they tracked PRN 22. OKEE is a good example of a station with poor GPS data quality. From many fragmentations of ionospheric delay estimates of OKEE (red), it is evident that carrier-phase measurements are corrupted by numerous cycle slips, short arcs, and outliers. OKEE (red) also held a small amount of data compared to AVCA (blue). Figure 4 shows the ionospheric spatial gradients between OKEE and AVCA. The leveling errors due to the short arcs are observable at each end of the curve, and the large gradients due to the excessive cycle slips are evident in the center of the curve. This example well illustrates that the poor data quality degrades the accuracy of ionospheric delay estimation and produce extremely large ionospheric gradients which are not real.

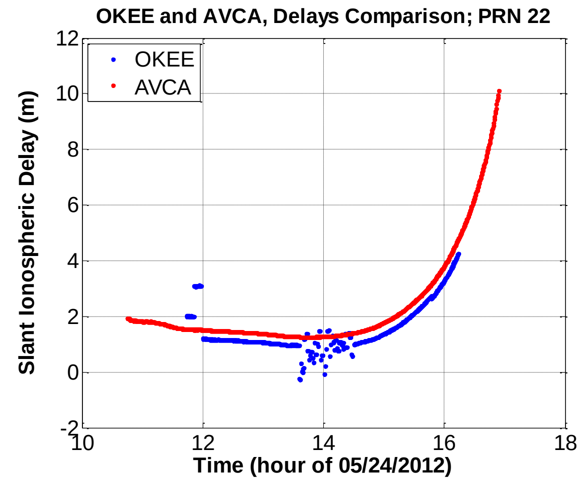

Figure 3. Dual-frequency Slant Ionospheric Delay Estimates for Stations OKEE (Poor Quality Data) and AVCA (Good Quality Data).

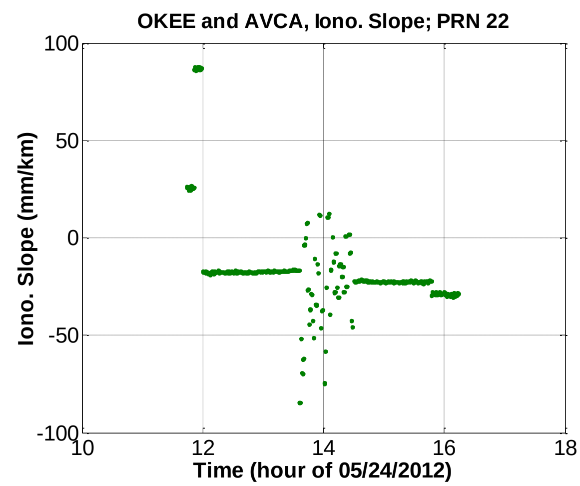

Figure 4. Ionospheric Gradient Estimates Corrupted by the Poor Quality Data of Station OKEE.

4.0. DATA ANALYSIS METHODOLOGY

The detailed methodology to select the station for LTIAM is composed of three steps: measuring GPS data quality information, determining thresholds of data quality parameters to remove poor-quality stations and selecting stations considering geographical condition. First, the data quality information is obtained by processing the RINEX file collected at each station through a series of data-quality-measurement algorithms. Then, the stations with poor data quality that need to be removed are selected based on whether the quality parameters of the station exceed the established thresholds. Data with higher sampling rates is preferred to observe ionospheric gradients accurately and to validate anomalous events using L1 code-minus-carrier measurements. Thus, these stations are ranked based on the data quality and data sampling rate. Third, to observe anomalous ionospheric gradients in CONUS, the selected stations should cover all of CONUS with separations of less than 40 km to the degree possible. To meet this criterion in regions with relatively few stations, some degree of data quality may need to be sacrificed. Thus, we examine the geographical contribution of stations selected to be removed in the order of the rank. Stations whose location increases the geographical observability of ionospheric behavior are restored despite their poor data quality. The details of each step are described in the following subsections.

4.1. DATA QUALITY MEASUREMENT ALGORITHMS

The input of the GPS data-quality measurement algorithms is the RINEX file collected from a station of our interest for two consecutive days and the output is the GPS data quality information of the corresponding station. As shown in Figure 5, these algorithms are composed of mainly three parts: LTIAM pre-processing, 'Translation, Editing, and Quality Check (TEQC)' algorithm, and adaptive filter algorithm.

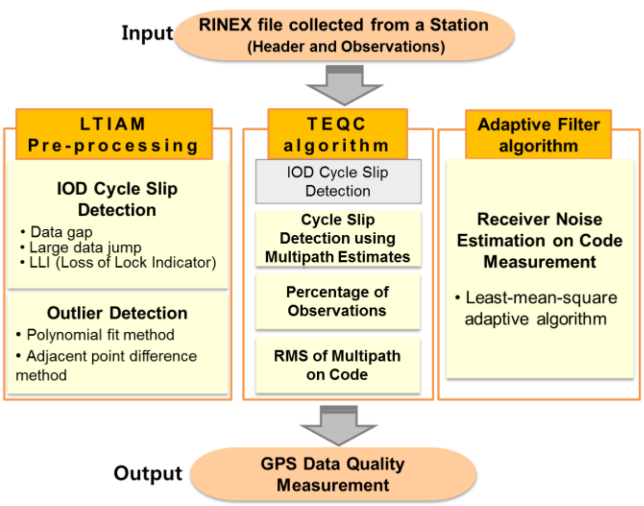

Figure 5. GPS Data-Quality-Measurement Algorithms.

Cycle slip and outlier detection methods have been already developed as a part of LTIAM pre-processing [26]. These detections are performed for each continuous arc of slant ionospheric delays estimated using dual-frequency carrier phase measurements. Three detection criteria (data gap, data jump and loss of lock indicator) are applied to identify IOnospheric Delay (IOD) cycle slips. After performing detection of cycle slips, outlier detection are carried out for each continuous arc. Two approaches, the polynomial fit method and the adjacent point difference method, are executed in parallel to detect outliers. LTIAM also detects short arcs, which are less than ten data points or five minutes, because leveling errors of those arcs are typically large and cause ionospheric delay estimation errors. The detailed methods are described in [26]. In this step, the number of IOD cycle slips, the number of outliers and the number of short arcs are counted as data-quality measurements.

Second, we implemented the measures of TEQC, which is developed by the University Navstar Consortium (UNAVCO) and is the most commonly used software to solve many pre-processing problems [24, 31]. The TEQC method is used to obtain quality information, which includes the percentage of observations, the mean of multipath on L1 code and L2 code, and the number of cycle slips detected using multipath estimates. The percentage of observations is the ratio of 'possible observation' to 'complete observation', where 'possible observation' indicates the total number of possible observation epochs in a given time window, and 'complete observation' is the number of epochs that actually observed code and phase data.

The IOD cycle slip detection of LTIAM performs better than that of TEQC by applying three detection criteria. However, the cycle slips occurred on both L1 and L2 simultaneously cannot be detected using IOD measurements. Thus, we upgrade the cycle slip detection by incorporating the method of TEQC which detects cycle slips using multipath (MP) estimates. The MP cycle slip is detected by using the linear combinations of the L1/L2 code (ρ_{L1}, ρ_{L2}) and carrier (ϕ_{L1}, ϕ_{L2}) measurements [24]. The linear combinations are defined as

$$
\begin{aligned}
MP1 &\equiv \rho_{L1} - \left(1+\frac{2}{\gamma-1}\right)\phi_{L1} + \left(\frac{2}{\gamma-1}\right)\phi_{L2} \\
&= M_{L1} + B_1 - \left(1+\frac{2}{\gamma-1}\right)m_{L1} + \left(\frac{2}{\gamma-1}\right)m_{L2} + \varepsilon_1 \\
MP2 &\equiv \rho_{L2} - \left(\frac{2\gamma}{\gamma-1}\right)\phi_{L1} + \left(\frac{2\gamma}{\gamma-1}-1\right)\phi_{L2} \\
&= M_{L2} + B_2 - \left(\frac{2\gamma}{\gamma-1}\right)m_{L1} + \left(\frac{2\gamma}{\gamma-1}-1\right)m_{L2} + \varepsilon_2
\end{aligned}
\tag{1}
$$

M_{Li} and m_{Li} are the multipath on code phase and carrier phase measurement of the Li frequency, respectively. The bias terms, B_1 and B_2, are

$$
\begin{aligned}
B_1 &\equiv -\left(1+\frac{2}{\gamma-1}\right)N_{L1} + \left(\frac{2}{\gamma-1}\right)N_{L2} \\
B_2 &\equiv -\left(\frac{2\gamma}{\gamma-1}\right)N_{L1} + \left(\frac{2\gamma}{\gamma-1}-1\right)N_{L2} \\
\gamma &= \frac{f_{L1}^2}{f_{L2}^2}
\end{aligned}
\tag{2}
$$

N_{Li} is the integer ambiguity of the Li frequency, and γ is the square of frequency ratio. When the data jump between two adjacent points in each continuous arcs of

MP1 and MP2 is greater than 10m, it is identified as a cycle slip. If the cycle slip occurs at a different point in time compared to the IOD cycle slip, this cycle slip is referred to as an MP slip.

After performing the cycle slip detection, the biases of the sub-arcs of MP1 and MP2 divided by the cycle slip are assumed to be a constant unless there is an undetected cycle slip remaining in the MP1 and MP2. Therefore, the constant is removed from each arc and the root mean squares (RMS) of these linear combinations are reported. Although the portion of phase multipath is included in this reported value, the amount is small compared to that of code multipath. Thus, the estimated RMS of MP1 and MP2 can be approximated to be the multipath on L1 code and L2 code, respectively [24].

Third, an adaptive filter algorithm is designed to estimate receiver noise on code measurements. After removing the bias components, B_1 and B_2 , of MP1 and MP2 from Equation (1), the MPi_new can be expressed as

$$MPi\_new = MP_i + \varepsilon_i \tag{3}$$

MP_i , the Li-frequency code multipath, is likely to be highly correlated to the MP_i of the consecutive previous day. However, ε_i , the receiver noise on Li code, is not correlated to the ε_i of the consecutive previous day. Therefore, the MPi_new of two consecutive days can be separated into the correlated component (MP_i) and the uncorrelated component (ε_i) using an adaptive filter [25]. The adaptive filter takes two inputs: primary input and reference input. In this study, the MPi_new of the day of interest is set as the primary input and the MPi_new of the previous day is set as the reference input. Then, the output of a Finite-duration Impulse Response (FIR) filter is calculated using the reference input and weights. The least-mean-square (LMS) algorithm has been used to adaptively adjust the weights of the FIR filter, which minimize the sum of squared errors of estimation [25].

The adaptive filter returns the part of the primary input which has strong correlation with the reference input as its output. Thus, the MP_i of the primary input, multipath estimate on code measurement, is calculated as the output of the adaptive filter. The estimation error of the filter represents the receiver noise on code, ε_i , because it represents the value with MP_i removed from the primary input. As explained, in order to estimate the receiver noise, ε_i , the correlation between the MPi of the two consecutive days has to exist. However, there are cases where such correlation is not clearly visible depending on receiver/antenna type and environmental changes. In these cases, the receiver noise in the quality output is presented as N/A.

4.2. THRESHOLDS OF DATA QUALITY PARAMETERS

Among the quality output of GPS data quality measurement algorithms we use seven parameters which affect the performance of LTIAM most. Those are the number of IOD cycle slips, the number of short arcs, the number of outliers, percentage of observation, multipath on L1 code and L2 code, Latency. Latency shows whether the RINEX files of the days tested are all loaded.

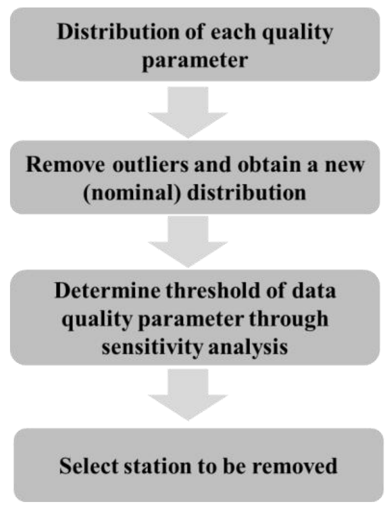

Figure 6. Steps to Select Stations to be Removed.

Figure 6 shows the steps to determine stations to be removed based on the seven quality parameters. We first collect data from CORS stations in CONUS for seven (or longer) consecutive days and obtain statistical distributions of each quality parameter. The data exceeding $\mu + 9\sigma$, mean plus nine times standard deviation, of the distribution is classified as extreme outliers. After discarding the outliers from the distribution, we obtain a nominal distribution of each parameter. Then using this new distribution, we determine a threshold of each data quality parameter through sensitivity analysis. As lowering the threshold, the $\mu + k\sigma$ of nominal distribution, both the number of stations removed, $M_{stations}$, and the number faulty ionospheric anomaly candidates removed, $N_{candidates}$, increase. Note that the faulty candidates are desired to be removed as many as possible. However, we should avoid removing too many good-quality stations which do not generate faulty anomalies while removing only a few more faulty candidates by lowering the threshold one step further. Thus we check the sensitivity ratio, λ , that measures the relationship between the number of faulty ionospheric gradients removed and the number of stations removed.

$$\lambda = \frac{N_{candidates}(k(i)) - N_{candidates}(k(i+1))}{M_{stations}(k(i)) - M_{stations}(k(i+1))};$$
$$where \quad i = 0,1,...,\max; \qquad (4)$$
$$k(0) = 9, \quad k(\max) = 0.5, \quad k(i+1) - k(i) = 0.1$$

Finally, the k value which returns the largest sensitivity ratio is chosen to set the threshold of each parameter. The same process is performed for all seven data quality parameters. If at least one parameter of a station exceeds its threshold, that station is removed from LTIAM processing. The example results of this process are shown in Section 5.

4.3. STATION SELECTION CONSIDERING GEOGRAPHICAL DENSITY CONDITION

The stations with poor data quality to be removed are determined through the data quality check as explained in the previous subsection. However, if a particular station increases the geographical observability of ionospheric behavior in a certain area with relatively few stations, it should be selected despite its poor data quality. Before performing geometry check on the station which are initially determined to be removed, the data quality and data sampling rate of each station is considered first to determine the rank of these stations. First, the rank is determined based on the data quality of each station. The number of parameters that exceed its threshold is counted and a greater number results in the station being designated as more undesirable. If the number of parameters that exceed its threshold is the same for two stations, the degree of excess over the threshold is measured and used to determine the rank. Once the undesired station rank is determined, it is reshuffled by taking into consideration the data sampling rate. CORS network stations provide data with different sampling rates of 1, 5, 10, 15, or 30 seconds. A faster data sampling rate is desirable. Therefore, remaining rank of stations with data sampling rates 1 second, stations with data sampling rates of 5 seconds, 15 seconds, and 20 seconds move upward in the ranking by 2 steps, 3 steps, and 10 steps, respectively.

Once the rank of stations within the poor-quality set to be removed is determined, a geometry check is conducted for each station. To finally decide whether to remove a station of poor data quality or not, we check the coverage of stations within a 100km radius of the corresponding station. The coverage of stations is defined as the area where pairs of stations whose baseline is less than 100km form. If a poor quality station to be removed has a station nearby, the change in coverage of stations will be small, even after the poor quality station is removed. Therefore, if the loss of coverage after discarding a station is more than 30% of original coverage, it is restored despite poor data quality. This geometry check is performed for each station to be removed in the order of the predetermined station ranking. Stations are removed one by one and the geometry check is iterated at each step. If a station is classified as a geographically critical station, even once when the two checks are conducted, the station is restored. This is done in order to leave a station with relatively acceptable quality among poor-quality stations in the case that stations with bad quality are concentrated in one area.

5.0. RESULTS OF CORS STATION SELECTION

The dates from which data were collected and analyzed to evaluate the performance of the GPS data quality check station selection algorithms are shown in Table 1. The geomagnetic conditions on the seven consecutive days are shown with two indices of global geomagnetic activity from space weather databases: planetary K (Kp) and disturbance, storm time (Dst). The number of operating stations of CORS network in CONUS is 1578 in this period.

Table 1. Dates Analyzed to Investigate Quality of Stations of CORS Network in CONUS

Day (UT mm/dd/yy)	Kp	Dst
24/05/12	2.0	-15
25/05/12	2.3	17
26/05/12	2.3	-6
27/05/12	1.3	14
28/05/12	2.3	23
29/05/12	2.3	23
30/05/12	2.3	16

As Kp and Dst in the Table 1 indicate, the geomagnetic storm condition was very quiet. Therefore, data qualities of the stations of CORS network in CONUS can be observed while minimize any influence of abnormal ionospheic behavior to the quality measurements. Also, since it is thought that an anomalous ionospheric event has not occurred in this period, the threat candidates that are results from LTIAM processing can be ascertained as faulty candidates generated by processing poor quality data. The results of the procedures for data quality determination and CORS station selection at each step are as follows.

오류! 참조 원본을 찾을 수 없습니다.2 shows the example results from the GPS data-quality measurement algorithms for station NVLA on 26 May 2012. The elevation cutoff angle is 10-degrees as a default. Among output parameters, seven parameters in color and latency are used, as determining quality of observation data to select the CORS stations. The number of IOD slips, the number of outliers, and the number of short arcs are counted using the method of LTIAM. The percentage of observations, the number of MP slips, mean MP1, and mean MP2 are measured from the method of TEQC.

Table 2. Data-Quality Information for Station NVLA on 26 May 2012

Output Parameters	Example	Description
Date	26 May 2012	Day Year
Station ID	NVLA	
Receiver type	LEICA GRX1200PRO	
Antenna type	LEIAT504	
Possible observation (> 10 deg)	25778	Total number of possible observation epochs in a given time window
Complete observation (> 10 deg)	25728	Number of epochs that actually had L1/L2 code and phase data from at least one SV.
Percentage of observations	100	(Complete observation / possible observation) x 100
Mean S1 (> 10 deg)	46.43	Mean signal to noise ratio (SNR) for L1
Mean S2 (> 10 deg)	42.27	Mean signal to noise ratio (SNR) for L2
IOD slips (> 10.0 deg)	51	Total number of ionospheric delay (IOD) slip occurred
MP slips (> 10.0 deg)	1	Total number of Multipath slip occurred
Outliers (> 10.0 deg)	11	Total number of outlier observed
Short arcs (> 10.0 deg)	38	Total number of short arc
Mean MP1 (> 10 deg)	0.2830 (m)	Mean of multipath on L1 code
Mean MP2 (> 10 deg)	0.3222 (m)	Mean of multipath on L2 code
Receiver noise1 (>10 deg)	0.1024 (m)	Mean of receiver noise on L1 code
Receiver noise2 (>10 deg)	0.0954 (m)	Mean of receiver noise on L2 code

Table 3. Rank of the Station of CORS network in CONUS; the Worst Case is on the Top for Each Quality Parameter

	# of IOD cycle slip		Per. of Obs.		# of Short arc		# of Outlier		Mean of MP1	
Rank	Stn.	#	Stn.	%	Stn.	#	Stn.	#	Stn.	meter
1	bru5	5552	p702	18	bru5	5545	mion	281.86	defi	0.7244
2	sag5	1544	p699	38.33	covx	1483.71	ls02	100.33	wach	0.718
3	covx	1529.43	ncwj	42.14	sag5	1466.43	frtg	67.71	ormd	0.7047
4	ls02	1301.5	twhl	50.71	ls02	1256.17	jxvl	65.57	zoa2	0.696
5	mlf5	1063	okee	59.71	mlf5	1051	okee	59.71	zfw1	0.6852
6	kns6	862.29	barn	61	kns6	862.14	cpac	57	zla1	0.6797
7	loz1	832.29	wvbr	61	kew6	819.57	pltk	55.29	zau1	0.6766
8	kew6	819.71	loz1	64.86	loz1	792.71	mipw	54.57	zob1	0.6461
9	okee	801.57	ohfa	67	okee	763.57	njcm	52	zlc1	0.6346
10	red6	767.57	sag6	67	red6	760.14	mihl	50.86	zab1	0.6337
11	mion	766.71	hgis	68.86	drv6	705.86	hruf	47.57	zmp1	0.6335
12	drv6	715	kysc	68.86	mion	697.57	napl	46.86	zse1	0.6331
13	lou6	673.57	arm3	70	lou6	646.71	brig	45.14	zoa1	0.6297
14	plo5	625.14	dqcy	71.14	det6	617.86	adri	44.43	red6	0.623
15	det6	621.71	hamm	71.14	plo5	615.57	brtw	43.29	zma1	0.6226
16	prry	598.29	negi	71.29	kew5	574.57	p671	41.14	loz1	0.6178

Table 3 shows the rank in order of bad quality for each quality parameter, and the same color indicates the same station. The worst stations will be triggered by multiple data-quality parameters as shown in Table 3. The Station OKEE among stations in color was introduced as an example of stations with poor GPS data quality in section 3.

The results of analyzing the quality parameters of all stations of the CORS network in CONUS show us how widely the performance of stations can vary. Figure 7a shows the total number of IOD cycle slips counted over all satellites during 24 hours on each station. The station ID is plotted along the x-axis, and the number of IOD slip is plotted along the y-axis. These numbers are counted for seven consecutive days. The mean value (blue) and the minimum value (red) among seven days are close together for the most of stations, and this indicates that the bad performance of stations persist for the extended period of time. From this test, 1.2 percent of stations had more than 500 IOD cycle slips per day, and more than 12 percent of stations had more than 50 IOD slips. Note that the mean value of all seven days over all stations is 37.98. As can be seen from **오류! 참조 원본을 찾을 수 없습니다.** to **오류! 참조 원본을 찾을 수 없습니다.**, each station in CONUS shows differing quality performances for the other quality parameters and not just IOD cycle slips. It can be observed that the performance is maintained for a certain period.

Once the data quality of stations is determined, the selection criteria for each quality parameter are set in order to select stations of poor quality. **오류! 참조 원본을 찾을 수 없습니다.** to **오류! 참조 원본을 찾을 수 없습니다.** show the PDF of each quality parameter. Data collected on each station for seven days. The red vertical line in **오류! 참조 원본을 찾을 수 없습니다.** refers to $\mu+9\sigma$ where data (blue points) exist continuously from 0 to this line. After the line ends, the continuity of data ceases to exist. Therefore, in this study, data that goes over this base line are removed as extreme outliers.

After removing the extreme outliers, the threshold of each quality parameter is determined using nominal distribution. An example is shown in Figure 5 which pertains to the number of IOD cycle slips among the quality parameters. As lowering the threshold that is the mean plus k sigma, both the number of stations removed and the number of faulty candidates removed increase. Note that the more faulty candidates we remove, the better it is, but we don′t want to remove too many stations while removing only a few more candidates. Thus, the K values which return the max sensitivity ratio are chosen to set the thresholds. For the case of IOD cycle slip, the resulting threshold is mean plus 1.4 sigma.

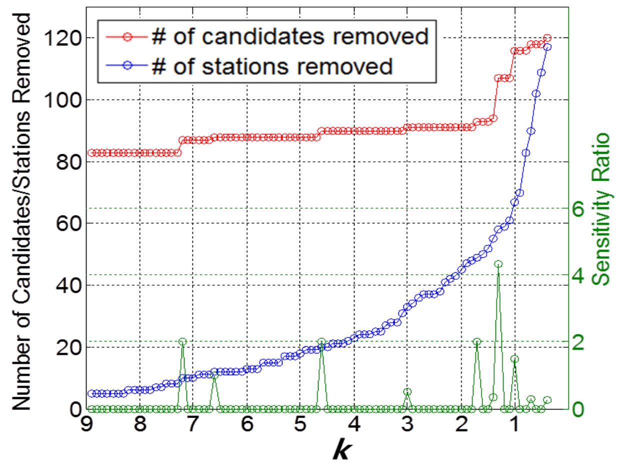

Figure 5. Comparison of number of candidates removed and stations removed and sensitivity ratio for IOD cycle slip.

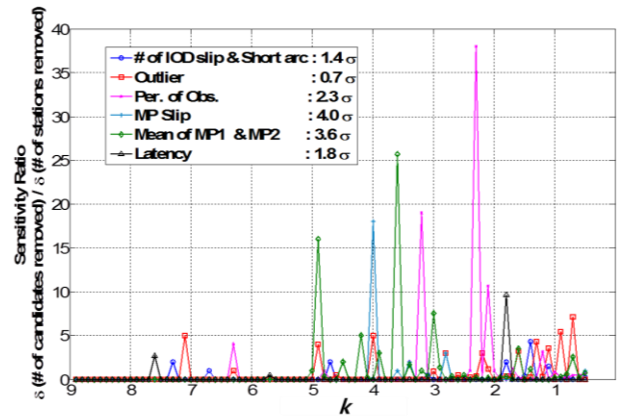

Figure 6 shows different sensitivity ratios with regards to the k value, which determines the reasonable threshold for all quality parameters. These sensitivity ratios are presented in different colors, and the label of

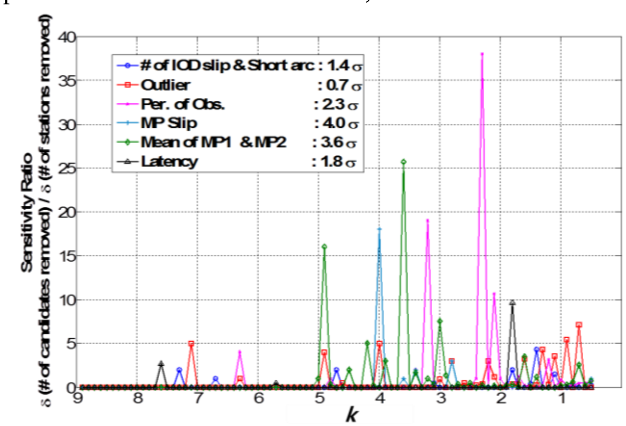

Figure 6 shows the k values that maximize the sensitivity ratio of each parameter.

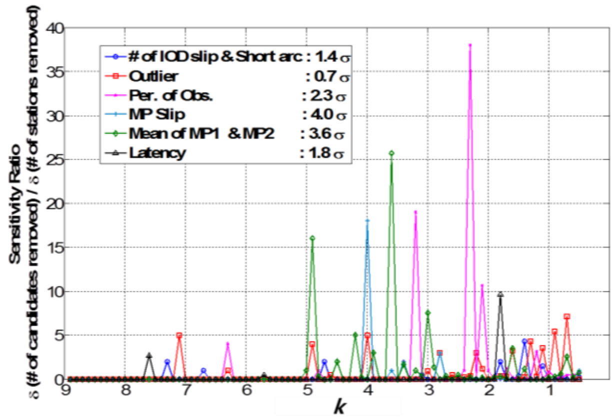

Figure 6. Sensitivity ratio for all quality parameters.

For each station in CONUS, each quality parameter that exceeds the set threshold is checked, and if one threshold is exceeded, the corresponding station is determined to be a station for removal. As shown in Figure 7, 308(19.4%) stations out of a total of 1587 operational stations were determined as stations to be removed due to poor quality in this study.

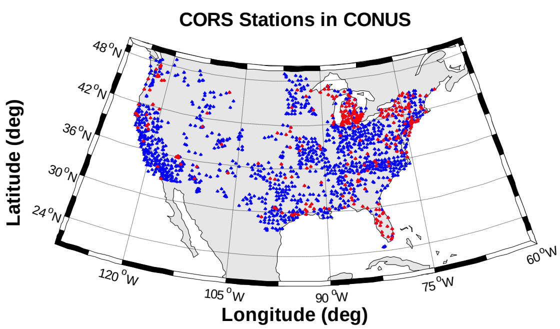

Figure 7. Map of CORS station in CONUS. The stations to be removed are in red and others are in blue.

As the result of LTIAM processing of May 26, 2012, 92 faulty candidates were generated. If the 308 stations selected from the quality check were removed, 81 faulty candidates would disappear, as shown in Table 2, and as a result, we can save time and effort spending to manually validate the faulty candidates to confirm that they are not real anomalies.

Table 2. Results of quality check on station of CORS network in CONUS on 26 May 2012.

# of stations removed in CONUS (out of 1587)	308 (19.4%)
# of faulty candidates removed at 05/26/2012 (out of 92)	81 (88.0%)

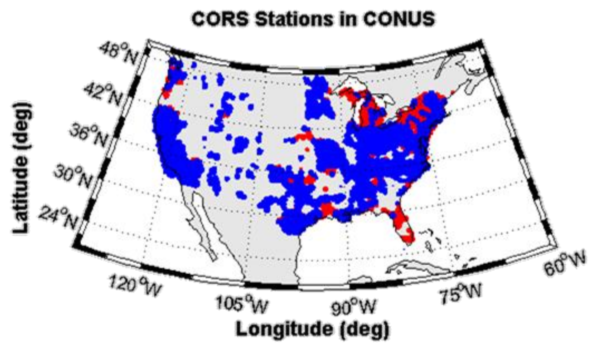

Figure 8. Loss of coverage (red) due to station removed.

Figure 8 shows the coverage of stations in CONUS when 308 stations were removed. Some stations, if removed, significantly reduce the coverage in certain areas. Therefore, if a particular station increases the geographical observability, we should keep it despite its poor data quality. The PRRY station shown in Figure 9 is a station that is restored as a result of geometry check. The region colored dark gray is the coverage of stations and Figure 9 shows that there is a large difference before and after the removal of the PRRY station. In the case of the PRRY station, the loss of coverage after the station was removed was approximately 80%. Likewise, in this study, if the loss of coverage is above 30%, the corresponding station is determined to be a geographically critical station, and despite being of bad quality, that station is not removed. In Figure 10, the loss of coverage that occurs when the PRRY station is removed is shown as a red colored region.

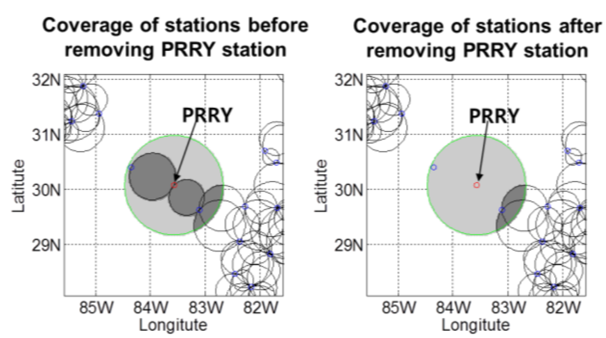

Figure 9. Comparison of coverage of stations before and after the removal of the PRRY station.

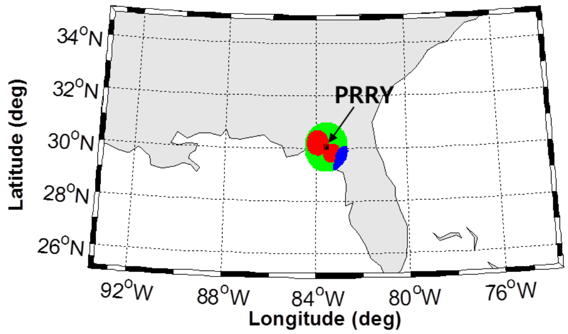

Figure 10. Map of loss of coverage (red) after removing PRRY station.

We performed the geometry check on 308 stations which were classified as bad stations initially. Among those, after geometry check, 56 stations were restored as geographically critical stations as shown Figure 11.

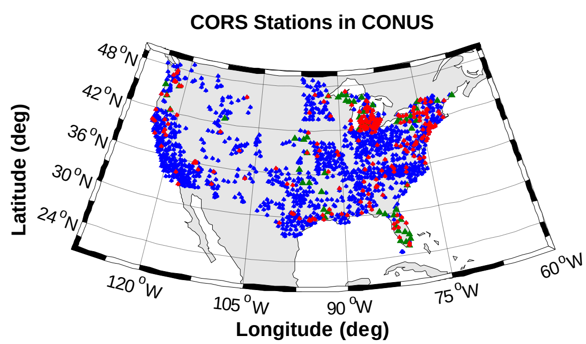

Figure 11. Map of CORS station in CONUS. The stations removed are in red and the stations restored are in green.

Referring to Table 3, the number of faulty ionospheric anomaly candidates removed on May 26, 2012, after the geometry check shows 81, which is equivalent to the number before geometry check. Therefore, in this case, 88% of false anomalies were removed while removing only16% of stations.

Table 3. Results of geometry check on station classified as bad stations on 26 May 2012.

	Before the geometry check	After the geometry check
# of stations removed in CONUS (out of 1587)	308 (19.4%)	252 (15.9%)
# of faulty candidates removed at 05/26/2012 (out of 92)	81 (88.0%)	81 (88.0%)

The second test was conducted on the 20 November 2003 ionosphric storm during which the largest gradients were observed to date. On 20 November 2003, the space weather indices, Kp of 8.7 and Dst of -472, indicate storm day. As a result of the LTIAM processing, there were a total of 25 candidates exceeding 300 mm/km ionospheric spatial gradient on this day, and the previous study reported that among them, 9 were true threat points. As shown in **오류! 참조 원본을 찾을 수 없습니다.** and **오류! 참조 원본을 찾을 수 없습니다.**, when 252 stations with poor data quality determined from quality check are removed, 9 faulty candidates are removed while 2 out of 16 true threat points are also removed. However, the removed threat points are not critical points that determine the boundary of the current threat model, and as the stations of the CORS network is dense to the present day, taking into consideration the geographical distribution through geometry check, even if actual true threat points have disappeared, equivalent ionospheric anomalous events will be able to be detected by different stations in near proximity.

6.0. UPDATED THREAT ANALYSIS FROM PAST STORM DATA

As explained above, the LTIAM is primarily intended to examine new CORS and IGS station data to detect recent and future anomalous ionospheric gradients. It also has the capability to re-examine past ionospheric storm data and discover new properties of it, as shown in [22]. This section continues the analysis in [22] to better understand the distribution of spatial gradients under anomalous ionospheric conditions. It then uses this analysis as a template for estimating anomalous gradient probabilities from future ionospheric data.

6.1. EXTENDED ANALYSIS OF 2000-2005 RESULTS

Table A (from [22]) shows the 10 days of known ionospheric storm activity in CONUS from 2000 to 2005 that were manually analyzed prior to the existence of LTIAM [2,3] and re-analyzed with LTIAM [22] to develop the current GBAS ionospheric threat model for CONUS. Figure A is one of the key results from [22] and shows the observed and validated anomalous ionospheric gradients over these 10 days. This includes gradients previously discovered from manual analysis and later re-confirmed by LTIAM (green triangles) and those first discovered more recently by LTIAM (blue diamonds). All of the observations shown in this figure were derived from the four days highlighted in yellow in Table 1. These four days, and especially 11/20/03, were caused by an especially strong coronal mass ejection from the Sun and represent the most severe storm days known to have occurred since GPS data has become available [2,3].

Table A. Storm Dates Analyzed to Develop CONUS Ionospheric Threat Model [22]

Day (UT mm/dd/yy)	Kp	Dst
04/06/00	8.3	-287
04/07/00	8.7	-288
07/15/00	9.0	-289
07/16/00	7.7	-301
09/07/02	7.3	-177
10/29/03	9.0	-350
10/30/03	9.0	-383
10/31/03	8.3	-307
11/20/03	8.7	-422
07/17/04	6.0	-76

The CONUS threat model limits shown in Figure A were chosen to bound the largest observed events. For this reason, previous data searches focused on identifying and validating the largest apparent gradients in the dataset. For automated LTIAM data analysis, this was done by limiting the outputs to apparent gradients above 200 mm/km (as measured by LTIAM from raw L1/L2 data). While some of the resulting validated gradients shown in Figure A are lower than this threshold, this is because validated gradient values are those that can be confirmed by comparing L1/L2 estimates with those from L1 code-minus-carrier analysis. When a discrepancy exists for an observation that appears to be valid, the minimum gradient that can be confirmed to be valid is reported, and many of these are under the original LTIAM reporting threshold. Therefore, the distribution of gradients in Figure A is roughly uniform (equally-distributed) between the minimum that represents anomalous conditions (25 – 50 mm/km) and the maximum of just over 400 mm/km. This does not appear to be a fair reflection of all anomalous conditions, in which we would expect smaller gradients to be much more frequent than larger ones.

To examine this further, the LTIAM was used to re-evaluate the 10 storm days in Table A with a lower threshold of 50 mm/km and an upper threshold of 200 mm/km. In other words, only gradients that pass the internal LTIAM checks and have estimated magnitudes between 50 and 200 mm/km were reported. This avoids any overlap between new observations and those made previously with a minimum threshold of 200 mm/km. Because of the large number of gradients discovered in this range, manual validation is not practical; thus the reported gradients are the "unvalidated" values from L1/L2 analysis and will tend to be larger than the equivalent “validated” ones for the reasons mentioned above.

Figure B shows the results in the same format as Figure A: gradient (or "Slope") vs. satellite elevation angle. A total of 2929 points were discovered between 50 and 200 mm/km. What is evident is that the number of gradients of smaller magnitudes greatly exceeds the number above 200 mm/km. In addition, between 50 and 200 mm/km, it is clear that smaller gradients are more likely. Figure C emphasizes this by showing the cumulative distribution of gradients between 50 and 200 mm/km (in other words, the *x*-axis probability of falling below the gradient indicated on the *y*-axis). Figure C shows that the median (50th percentile) of these gradients is 82.8 mm/km, while the midpoint of the range from 50 to 200 mm/km is 125 mm/km. Similarly, the 90th percentile is 149.1 mm/km, while the 90th percentile of the range is 185 mm/km.

The key unknown in evaluating the distribution of anomalous gradients is the fraction of the 2929 observations that are valid; i.e., they represent actual ionospheric gradients. We know from experience with the LTIAM is that lower apparent gradients are more likely to be valid than larger ones. The reason for this is that actual ionospheric gradients are limited by physics, while "false" anomalies due to receiver or database errors are not limited in this way and can take almost any size. Therefore, it is almost certain that the percentage of valid observations from 50 to 200 mm/km is higher than the same percentage above 200 mm/km. This percentage is about 30%, based on 73 validated measurements above 200 mm/km out of 243 outputs generated by LTIAM [22]. Note that Figure A shows a total of 99 validated measurements, but 26 of these resulted in validated events below 200 mm/km, and these are counted as being in the 50 – 200 mm/km range. As expected, a slightly smaller percentage applies for an LTIAM threshold of 300 mm/km (13 of 53, or about 24.5%).

If we use 30% as a lower bound on the percentage of valid observations between 50 and 200 mm/km, the resulting estimate is 2929 × 0.3 = 878.7. Adding the 26 previously validated observations in this range gives a total of 904.7, or about 905 valid observations. The 73 validated observations above 200 mm/km thus represent a fraction of 73/(905+73), or about 7.5% of the total set of (estimated) validated observations. Above 300 mm/km, the ratio is 13/(905+73), or about 1.3% of the total set. If we instead assume as an upper bound that all 2929 measurements between 50 and 200 mm/km are valid, the ratio of validated observations above 200 mm/km would drop to 73/(2929+26+73), or about 2.4%, while the ratio above 300 mm/km would drop to 13/(2929+26+73) , or about 0.4%.

These bounds show that, with very high confidence, the fraction of anomalous gradients above 200 mm/km in the 2000-2005 CONUS ionospheric storm data set is very small: below 10% at least, and very likely below 5%. Furthermore, since the distribution of LTIAM measurements between 50 and 200 mm/km is weighted toward the lower end, and lower gradients are more likely to be valid than higher ones, it is safe to conclude that the distribution of gradients throughout the CONUS threat space is heavily weighted toward the lower end of the gradient range. While GBAS ground stations must conservatively protect the entire threat space, including the highest possible gradients, this knowledge is not directly applicable to the ionospheric threat mitigations described in [1,4,5]. However, it helps us better understand ionospheric behavior relevant to GBAS during anomalous conditions, and it should lead to improved and less-conservative mitigation strategies in the future.

6.2. UPDATING PROBABILITY ESTIMATES WITH FUTURE DATA

The primary objective of LTIAM is to continually improve our understanding of both nominal and anomalous ionospheric gradient behavior. The re-analysis of past data in Section 6.1 shows an example of what can be learned. Going forward, the processing power of the LTIAM software allows automated analysis of all days in a particular region or a focus on particular days that appear anomalous based on external information such as ionospheric weather parameters [6,22]. The information collected over time from these results will allow us to update the probabilities estimated in Section 6.1 and gain a better idea of the "prior probability" of extreme ionospheric gradients (e.g., over 200 mm/km) (a) over all ionospheric conditions; and (b) conditional on the knowledge that anomalous conditions are present. It is important to distinguish these two results. If LTIAM processing of all data in a particular region is carried out, the distinction between "nominal" and "anomalous behavior" can be determined after the fact based on both the observed gradients and external space weather information. In this case, probability estimates under all conditions and under anomalous conditions can be computed separately. If LTIAM processing is only carried out for days thought to be anomalous based on external information, only the latter estimate can be made.

Updating the results in Section 6.1 is mostly a process of repeating the same analysis procedure with future data while including data already analyzed from the past. The desired result is a data-driven distribution of anomalous gradients, meaning gradients above 25 or 50 mm/km. The lower number represents the lower limit of the anomalous gradient space in the CONUS threat model, but gradients at this level are not threatening to GBAS. Therefore, it may be more practical to limit LTIAM searches to 50 mm/km or more, as done in this paper. While anomalous gradients of any magnitude are rare, continual monitoring will occasionally discover them and allow new points to be added to the observations made to date. Thus, our knowledge of the distribution of anomalous gradients will grow slowly with time.

The rarity of severely anomalous gradients makes estimating their prior probability difficult. The model proposed in [23] uses the information available in early 2006 to estimate the probability of days with extremely anomalous ionospheric behavior, defined as days with gradients above 200 mm/km that could threaten GBAS (as shown in Table A, all such days in the 2000-2005 database had Kp indices greater than 8.0). This probability was estimated as a mean of 0.00196 and a 60^{th} percentile upper bound of 0.00257.

The ability of LTIAM to process all days (or at least all days with significant ionospheric activity) allows us to improve upon the precision and usefulness of these numbers. Given a set of days analyzed (either "all days" or "days of significant activity"), the first probability to be computed is the fraction of days with any anomalous gradients, defined as either above 25 or above 50 mm/km. Combining this result with the distribution of gradients above 25 or 50 mm/km allows the computation of the probability of observing gradients above any particular magnitude (e.g., 200 mm/km) per unit time. While "days" was the time interval used in [23], LTIAM outputs can and should be grouped into smaller intervals, likely "hours," to better reflect changing ionospheric conditions over the 24-hour "daily" cycle as well as the relatively short duration of anomalous gradients affecting particular areas.

7.0. SUMMARY

This paper provides an overview of the LTIAM software processing tool and demonstrates how it is used to identify potential ionospheric spatial anomalies from raw data collected by existing networks of CORS and IGS stations. Although LTIAM does its own screening of the raw data, significant ionospheric anomalies are rare; thus most of the results from LTIAM are "false" anomalies created by receiver or database errors. The vast majority of "false" anomalies come from relatively small subsets of CORS and IGS stations with poor data quality. The data quality evaluation methodology developed in this paper successfully reduced the number of false anomalies by almost 90% by removing only 16% of the CORS stations in CONUS. This was achieved while retaining stations with marginal data quality in geographically key locations. The end result is that the number of outputs requiring manual validation is greatly reduced.

This paper also re-examines the database of known ionospheric storm events in CONUS from 2000 to 2005 to estimate the distribution of anomalous gradient magnitudes. Examining all gradients estimated to be greater than 50 mm/km by the LTIAM shows that the validated events with gradients above 200 mm/km are greatly outnumbered by events with gradients from 50 to

100 mm/km. The degree to which gradients above 200 mm/km are unusual depends upon the percentage of valid events from 50 to 200 mm/km, which is almost certainly larger than the approximately 30% that applies to anomalies above 200 mm/km in the same dataset. A simplified method is proposed to update this probability and estimate the overall probability of anomalous ionospheric conditions as more data is collected over time.

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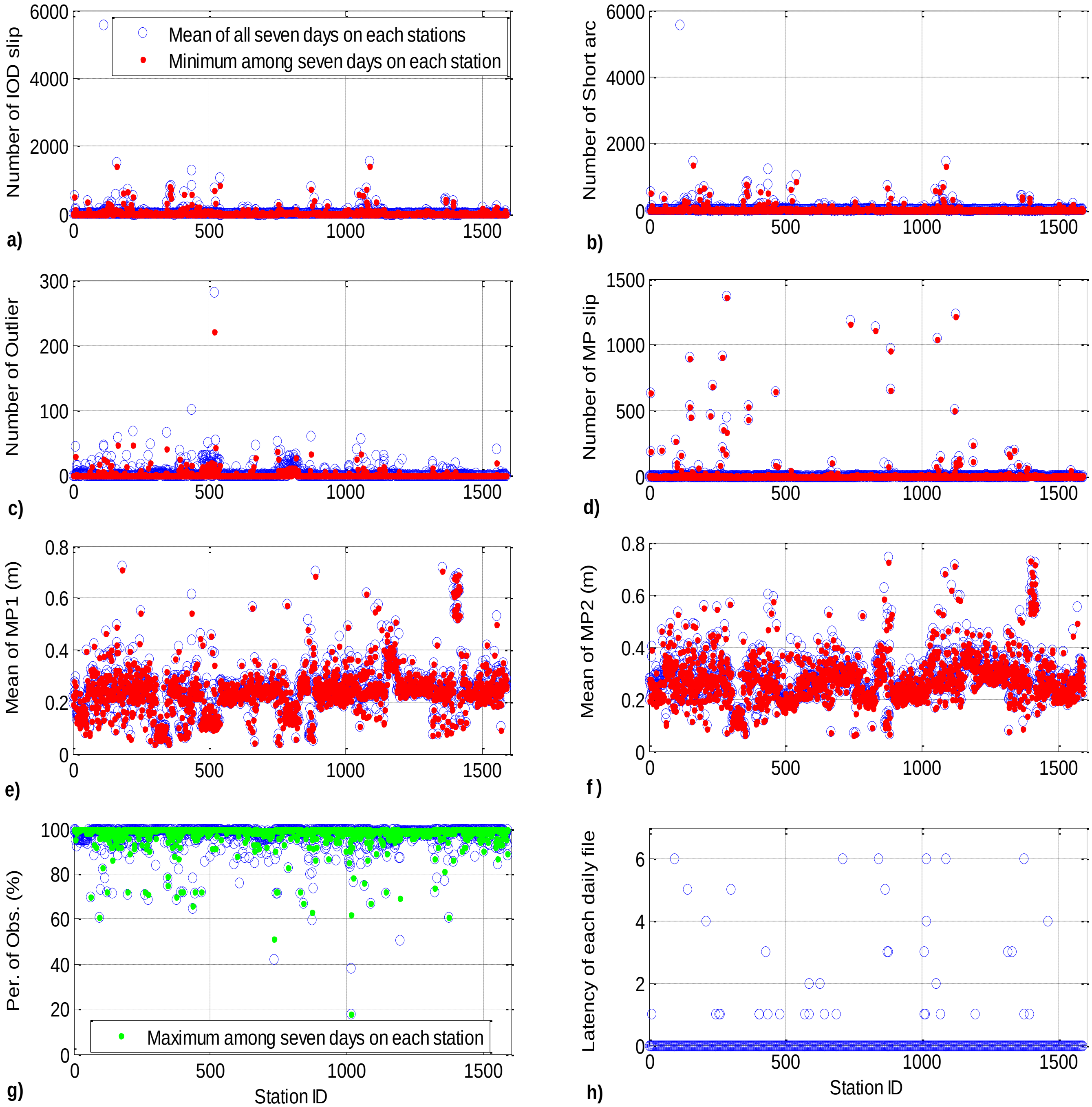

Figure 12. Quality parameters occurred on each station per day: a) number of IOD cycle slips which mean IOD cycle slips of all 7 days and all stations is 37.98, b) number of short arcs which mean short arcs of all 7 days and all stations is 32.74, c) number of outliers which mean outliers of all 7 days and all stations is 3.14, d) number of MP slips which mean MP slips of all 7 days and all stations is 13.24, e) mean of MP1 which average of mean MP1 of all 7 days and all stations is 0.2457 m), f) mean of MP2 which average of mean MP2 of all 7 days and all stations is 0.2826 m, g) percentage of observations which mean percentage of observations of all 7 days and all stations is 97.19%, and h) latency of each daily file for 7days.

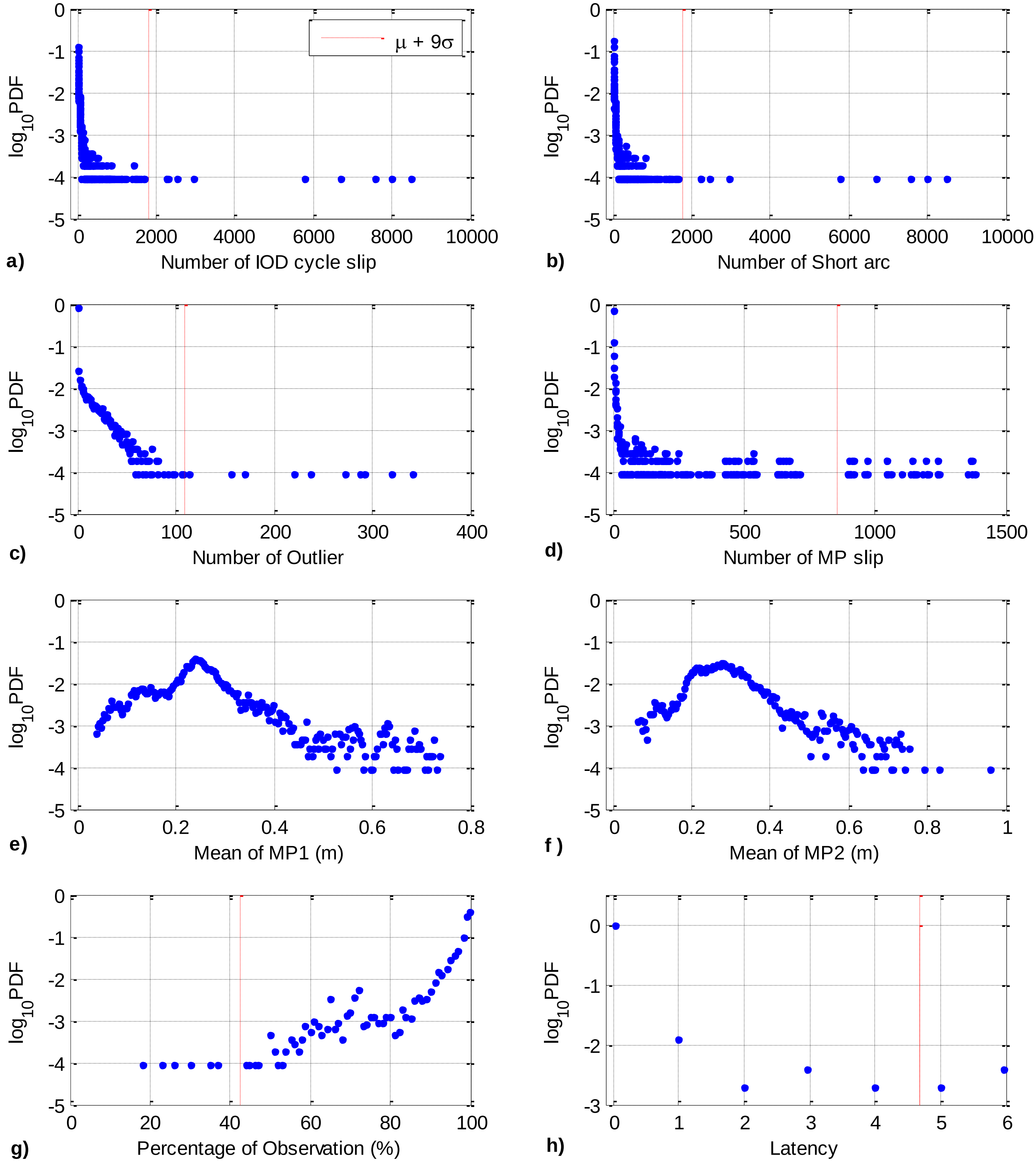

Figure 13. Probability density function of the quality parameters on each station per day, data collected for 7 days: a) number of IOD cycle slips, b) number of short arcs, c) number of outliers, d) number of MP slips, e) mean of MP1, f) mean of MP2, g) percentage of observation, and h) latency.

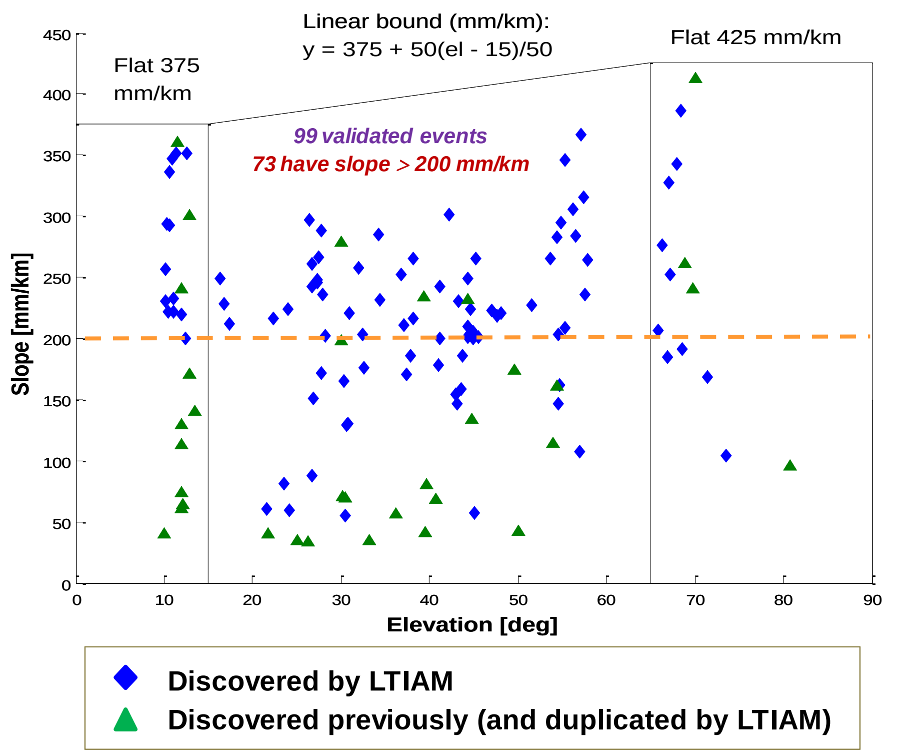

Figure A. CONUS Ionospheric Threat Model with Observed Spatial Gradients (LTIAM Threshold = 200 mm/km)

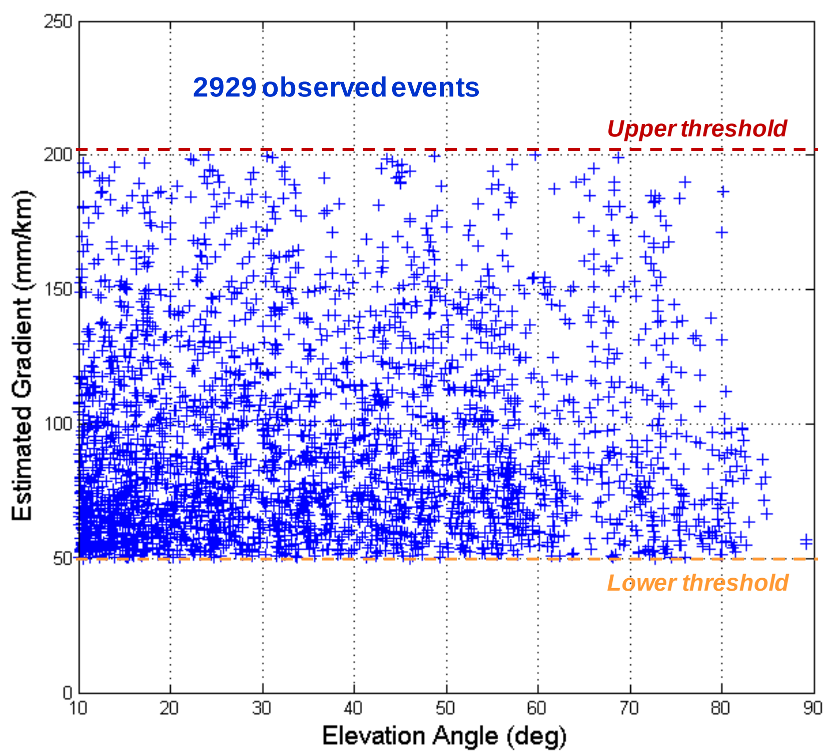

Figure B. CONUS Ionospheric Gradients from LTIAM (50 to 200 mm/km)

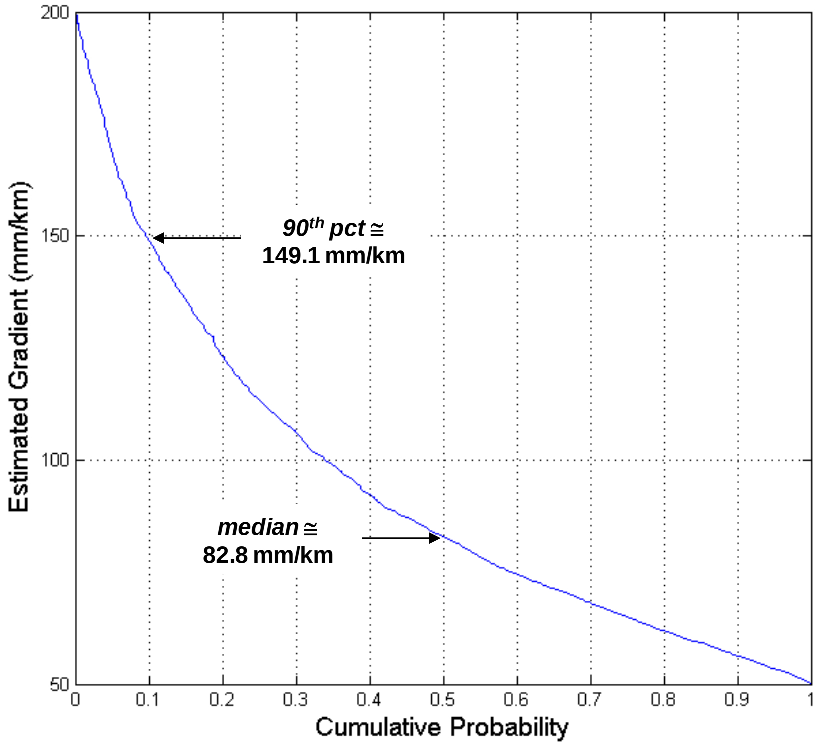

Figure C. Cumulative Distribution of CONUS Ionospheric Gradients from LTIAM (50 to 200 mm/km)